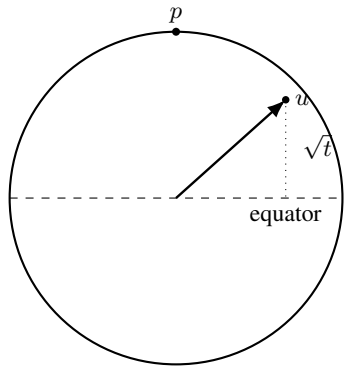




Warmup picture. If a rule depends only on the squared height $t = (u \cdot p)^2$, then an orthonormal triple gives three numbers a, b, c with $a + b + c = 1$. The proof becomes a one-variable problem on $[0, 1]$.